

Lab 3 - Written Architecture Justification

Chosen architecture: Layered Architecture.

Reason 1 - Separation of concerns: citizen-facing functions, scheduling and vaccination operations, and persistent data are kept in distinct layers/components.

Reason 2 - Controlled data access: vaccination data is accessed through defined interfaces, while authentication and access control sit at the service boundary.

Security advantage: this provides a clear boundary for authentication and authorisation before sensitive citizen or officer information is accessed, supporting Lab 1 NFR-002.

Performance benefit: the certificate and QR verification path can be kept focused and measured against Lab 1 NFR-001, which specifies verification in under 150 ms.

This Lab 3 continues the Vaccination Cohort & Dose Scheduling System defined in Lab 1.